COMP102 Lecture 7: Binary Heaps

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Plan

1 Learning Outcomes
2 From Queues to Priority Queues
3 Heap Property
4 Array Representation
5 Insertion and Removal
6 Python heapq
7 Priority Scheduling
8 Complexity and Limits

Learning Outcomes

Learning outcomes

By the end of this lecture, students should be able to:

- define the **priority queue** abstract data type;
- distinguish a normal queue from a priority queue;
- define the **min-heap** and **max-heap** properties;
- explain why a heap is **not** a binary search tree;
- represent a binary heap using a Python list;
- trace insertion using **sift up**;
- trace removal using **sift down**;
- use Python's `heapq` module to implement priority queues;
- model a small **priority scheduling** problem;
- analyze the time complexity of heap operations.

Last lecture: BSTs used an ordering invariant to search for a specific value. This lecture uses a weaker invariant to repeatedly extract the next most important value.

From Queues to Priority Queues

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Normal queue

First in, first out.

The first task that arrives is handled first.

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Print homework	1	3
System update	2	2
Emergency shutdown	3	10

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FIFO answers: **who came first?** A priority queue answers: **who matters most right now?**

Priority queue intuition

Priority queue intuition

Queue

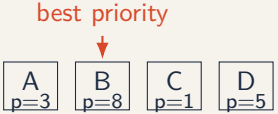
← A B C D

arrival order

Priority queue intuition

Queue

Priority queue



Priority queue intuition

Queue

← A B C D

arrival order

Priority queue

best priority

↓

A p=3 B p=8 C p=1 D p=5

A priority queue does not process items by position. It processes items by priority.

Priority queue as an ADT

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Abstract data type

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Operation	Meaning
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`pop()`	remove and return the best item
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The ADT says *what* behavior we want. We need a data structure to implement it efficiently.

Could we just use a list?

Implementation	Push	Pop best	Issue
Unsorted list	$O(1)$	$O(n)$	finding the best item is expensive
Sorted list	$O(n)$	$O(1)$	insertion is expensive
Binary heap	$O(\log n)$	$O(\log n)$	balanced compromise

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A heap is not magic. It is a compromise: both insertion and removal stay efficient.

Heap Property

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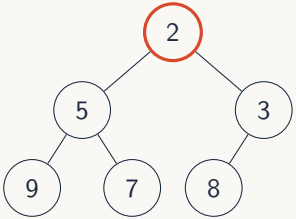

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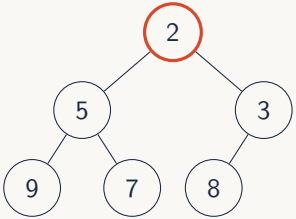

The heap property gives access to the best item. The shape property keeps the height small.

The min-heap property

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For every node:

$$\text{parent value} \leq \text{child values}$$

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Remark 3.3. Consequence

The smallest value is always at the root.

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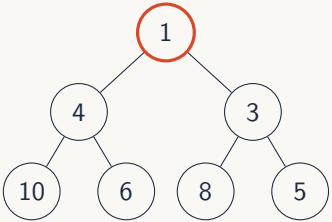

Min-heap and max-heap

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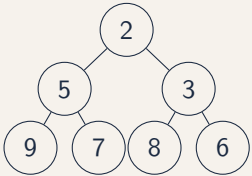
Min-heap

Parent $\leq$ children



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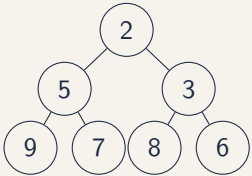

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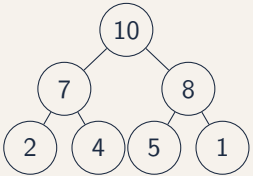

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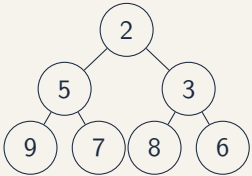

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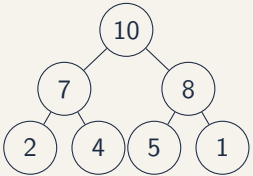

Same idea, opposite priority convention. In this lecture we mostly use min-heaps.

A heap is not a BST

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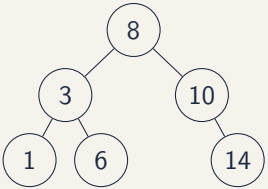
BST rule

left subtree $<$ node $<$ right subtree



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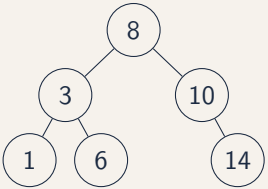

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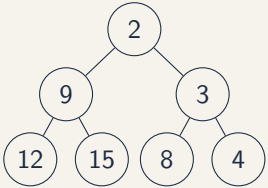

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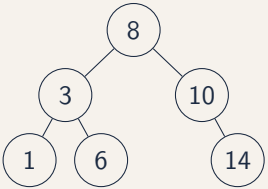

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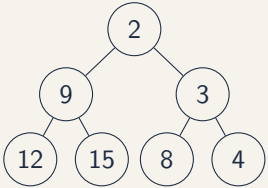
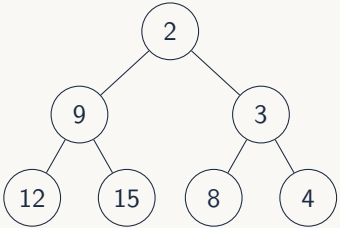


A heap only orders parents before children. It does not order the left subtree against the right subtree.

Valid heap but invalid BST



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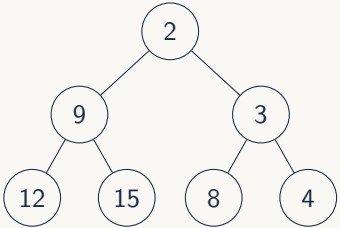

As a min-heap

Every parent is smaller than its children.

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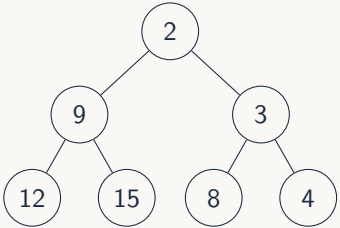

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The value 9 is in the left subtree of 2, but 9 is greater than 2.

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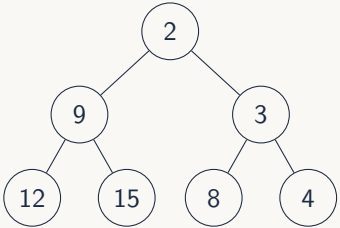

As a min-heap

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As a BST

The value 9 is in the left subtree of 2, but 9 is greater than 2.

This tree is a valid min-heap and an invalid BST. The two invariants are different.

The complete-tree shape

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Complete binary tree

All levels are full except possibly the last, and the last level is filled from left to right.

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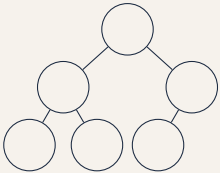

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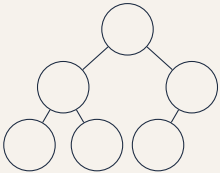

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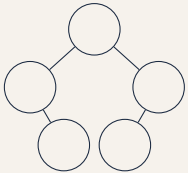

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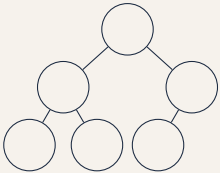

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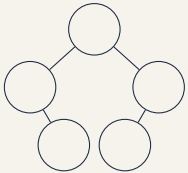

The shape rule prevents the heap from degenerating into a long chain.

Array Representation

A binary heap is stored as a list

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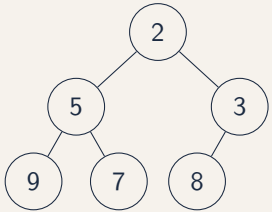
Tree view



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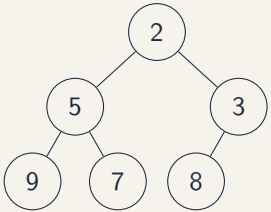


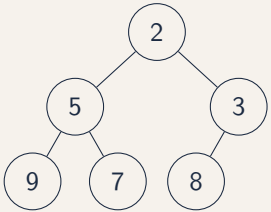
List view

2	5	3	9	7	8
0	1	2	3	4	5

`[2, 5, 3, 9, 7, 8]`

A binary heap is stored as a list

Tree view



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The list stores the complete tree level by level, from left to right.

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For a node at index `i`

Relation	Index
Left child	`2*i + 1`
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Remark 4.4. Important

This is why binary heaps usually do not need a `Node` class.

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A complete-tree shape gives us a compact array representation.

Checking the heap property in a list

For a min-heap, every valid parent index `i` must satisfy:

$$\texttt{heap[i]} \leq \texttt{heap[2*i + 1]} \quad \text{and} \quad \texttt{heap[i]} \leq \texttt{heap[2*i + 2]}$$

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[2, 5, 3, 9, 7, 8]
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Only parent-child comparisons matter. The list does not need to be sorted.

Insertion and Removal

Insertion: append, then repair upward

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Insertion only repairs one path: from the new leaf toward the root.

Trace: inserting 1

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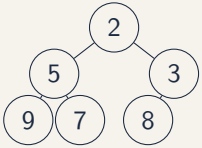

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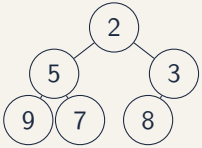

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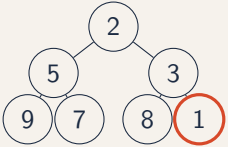

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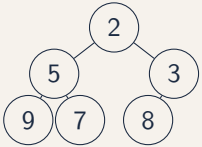

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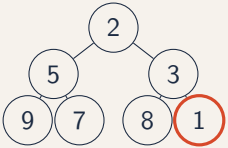

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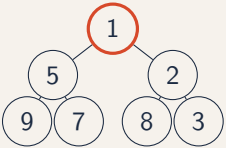
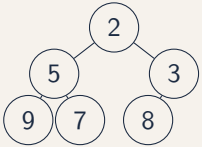
Sift up



[1,5,2,9,7,8,3]

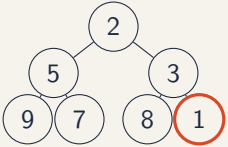
Trace: inserting 1

Start



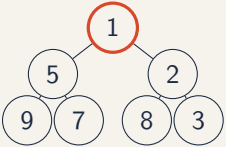
`[2,5,3,9,7,8]`

Append



`[2,5,3,9,7,8,1]`

Sift up



`[1,5,2,9,7,8,3]`

The new value climbs only while it is smaller than its parent.

Sift up in Python

```python
def heappush(heap, value):
    heap.append(value)
    i = len(heap) - 1

    while i > 0:
        parent = (i - 1) // 2
        if heap[parent] <= heap[i]:
            break
        heap[parent], heap[i] = heap[i], heap[parent]
        i = parent
```

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```

The loop moves upward through parent indexes, so it performs at most one swap per tree level.

Removing the minimum

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1. Save the root: this is the minimum value.

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Name of the repair
sift down

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Name of the repair

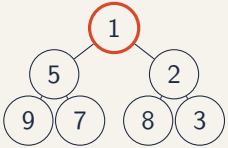
sift down

Removal is harder than insertion because the root disappears and the heap must choose the next best item.

Trace: popping the minimum

Trace: popping the minimum

Start



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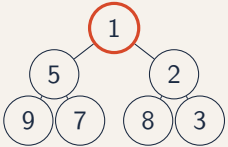
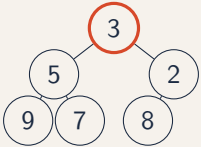


`[1,5,2,9,7,8,3]`

Move last to root



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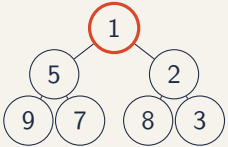

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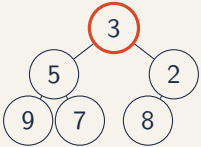

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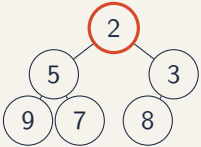
Sift down



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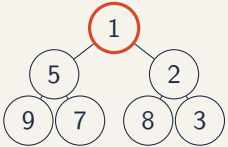

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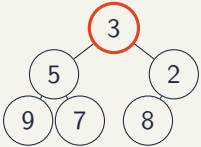

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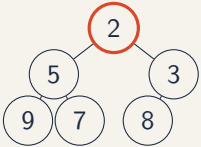

`[2,5,3,9,7,8]`

The replacement value moves downward until both children are greater than or equal to it.

Sift down in Python

```python
def heappop(heap):
    result = heap[0]
    heap[0] = heap[-1]
    heap.pop()

    i = 0
    n = len(heap)
    while True:
        left = 2*i + 1
        right = 2*i + 2
        smallest = i

        if left < n and heap[left] < heap[smallest]:
            smallest = left
        if right < n and heap[right] < heap[smallest]:
            smallest = right
        if smallest == i:
            break

        heap[i], heap[smallest] = heap[smallest], heap[i]
        i = smallest

    return result
```

Why operations are logarithmic

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Complete shape

A heap with n elements has height:

$$O(\log n)$$

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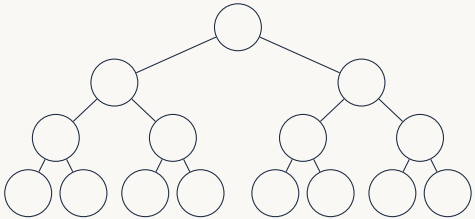

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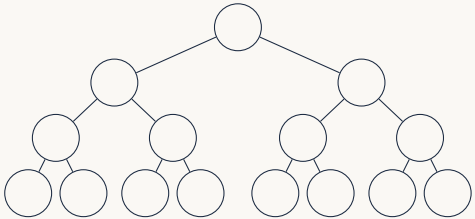

The heap never searches the whole tree during push or pop. It repairs one vertical path.

Python heapq

Python's heapq module

```python
import heapq

heap = []

heapq.heappush(heap, 5)
heapq.heappush(heap, 2)
heapq.heappush(heap, 8)

print(heap[0])              # 2
print(heapq.heappop(heap))  # 2
print(heap)                 # still a heap
```

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print(heap[0])              # 2
print(heapq.heappop(heap))  # 2
print(heap)                 # still a heap
```

heapq works on normal Python lists. It does not create a special heap object.

The internal list is not sorted

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A possible heap list

```
heap = [2, 5, 3, 9, 7, 8]
```

The internal list is not sorted

A possible heap list

```
heap = [2, 5, 3, 9, 7, 8]
```

This is a valid min-heap.

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heap = [2, 5, 3, 9, 7, 8]
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This is a valid min-heap.

But it is not a sorted list.

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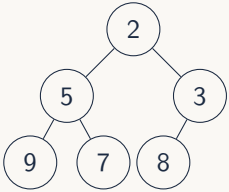

The internal list is not sorted

A possible heap list

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heap = [2, 5, 3, 9, 7, 8]
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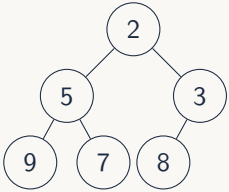

This is a valid min-heap.

But it is not a sorted list.

A heap keeps only enough order to make the next pop efficient.

Heapify

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Repeated insertion

```
heap = []
for x in data:
    heapq.heappush(heap, x)
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In-place heap construction

```
heapq.heapify(data)
```

Cost: $O(n)$

`heapify` transforms an existing list into a heap without first creating an empty heap.

Priority items with tuples

```python
import heapq

tasks = []

heapq.heappush(tasks, (3, "write report"))
heapq.heappush(tasks, (1, "fix server"))
heapq.heappush(tasks, (2, "answer email"))

while tasks:
    priority, task = heapq.heappop(tasks)
    print(task)
```

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while tasks:
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```

Tuples are compared from left to right, so the priority value controls the heap order.

Output order

For this heap:

```
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fix server
answer email
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The output is:

Result

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fix server
answer email
write report
```

In a min-heap, smaller numeric priority comes out first.

Tie-breaking equal priorities

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Risky pattern

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heapq.heappush(tasks, (1, task_a))
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Safer pattern

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```

The counter breaks ties by insertion order.

A common priority-queue entry is `(priority, counter, task)`.

When larger priority means more urgent

Python's `heapq` is min-heap oriented.

```python
import heapq

jobs = []

heapq.heappush(jobs, (-10, "emergency"))
heapq.heappush(jobs, (-2, "normal"))

priority, job = heapq.heappop(jobs)
print(job)  # emergency
```

When larger priority means more urgent

Python's `heapq` is min-heap oriented.

If larger numbers should come first, store negative priorities.

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```

Negating priorities turns a max-priority problem into a min-heap problem.

Priority Scheduling

Priority scheduling problem

We receive jobs with urgency levels.

Job	**Priority**
backup files	3
restart server	1
security alert	0
clean logs	5

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Lower number means more urgent.

Priority scheduling problem

We receive jobs with urgency levels.

Job	Priority
backup files	3
restart server	1
security alert	0
clean logs	5

Lower number means more urgent.

The scheduler should repeatedly select the job with the smallest priority number.

Priority scheduler with heapq

```python
import heapq

jobs = [
    (3, "backup files"),
    (1, "restart server"),
    (0, "security alert"),
    (5, "clean logs"),
]

heapq.heapify(jobs)

while jobs:
    priority, job = heapq.heappop(jobs)
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heapq.heapify(jobs)

while jobs:
    priority, job = heapq.heappop(jobs)
    print(job)
```

The heap stores pending jobs. Each pop chooses the next job to execute.

Scheduling with stable tie-breaking

```python
import heapq

jobs = []
counter = 0

def add_job(priority, name):
    global counter
    heapq.heappush(jobs, (priority, counter, name))
    counter += 1

add_job(1, "restart server")
add_job(1, "answer admin email")
add_job(0, "security alert")

while jobs:
    priority, order, name = heapq.heappop(jobs)
    print(name)
```

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When priorities are equal, the counter makes the scheduler behave predictably.

Where priority queues appear

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Scheduling

Operating systems, task managers, printers, emergency systems.

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Process the next event by earliest timestamp.

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Dijkstra's algorithm and Prim's algorithm need repeated extraction of the best candidate.

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Data analysis

Keep the top k items without sorting everything.

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Data analysis

Keep the top k items without sorting everything.

Priority queues are useful whenever *next best* matters more than full sorting.

Complexity and Limits

Complexity summary

Operation	Cost	Reason
`heap[0]` / peek	$O(1)$	best item is at the root
`heappush`	$O(\log n)$	sift up one path
`heappop`	$O(\log n)$	sift down one path
`heapify`	$O(n)$	build heap in place
Pop all items	$O(n \log n)$	n removals

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A heap gives fast access to the next best item, not fast access to every item.

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- searching for a specific value;
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- removing an arbitrary internal value;
- updating a priority already inside the heap;
- answering general order queries such as "what is the second smallest?" without extra reasoning.

A heap is specialized. It is not a replacement for BSTs, sorted arrays, or hash tables.

Choosing the right structure

Structure	Rule	Best for
Queue	first in, first out	arrival order
Stack	last in, first out	undo, backtracking
BST	left $<$ node $<$ right	searching by value
AVL tree	BST $+$ balance	reliable search
Heap	parent better than children	repeated best-item removal
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The right data structure depends on the question you need to answer quickly.

Wrap-up

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- A heap is not a BST: it has a weaker ordering rule.
- The complete shape allows compact list storage.
- Insertion uses sift up; removal uses sift down.

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- A priority queue removes items by priority, not arrival order.
- A binary heap is a complete binary tree with a heap property.
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- The complete shape allows compact list storage.
- Insertion uses sift up; removal uses sift down.
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A heap does not keep everything sorted; it keeps just enough order to find the next item efficiently.

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Dijkstra's algorithm repeatedly selects the unvisited node with the smallest current distance.

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Remark 9.3. Connection

That selection step is exactly a priority-queue operation.

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Remark 9.4. Connection

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Heaps are small structures with large consequences: scheduling, simulations, shortest paths, and optimization all rely on them.